

Pneumonia Detection Using a Custom Convolutional Neural Network with Hyperparameter Optimisation

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Abstract—This paper presents the development and evaluation of a Custom Convolutional Neural Network (CNN) for pneumonia detection using chest X-ray images.

Keywords—Pneumonia Detection, Convolutional Neural Network, Deep Learning, Hyperparameter Tuning, Medical Imaging

I. INTRODUCTION

Pneumonia is a serious respiratory disease that can be diagnosed through chest X-ray imaging. Manual diagnosis requires expert radiologists and may be time-consuming. Deep learning, particularly Convolutional Neural Networks (CNNs), has shown strong performance in image classification tasks, including medical imaging.

This study proposes a Custom CNN architecture for binary classification of chest X-ray images into:

- NORMAL
- PNEUMONIA

The primary objective is to evaluate how different hyperparameter configurations influence training performance and classification accuracy.

II. DATASET PREPARATION.

The experiments in this work utilized the Chest X-Ray Pneumonia dataset from Kaggle [<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>], which contains labeled X-ray images of NORMAL and PNEUMONIA cases. The dataset was organized into three subsets:

- Training Set
- Validation Set
- Test Set

A. Image Preprocessing

All images were resized to 224×224 pixels so that they match the input size required by the CNN model.

The pixel values were then normalized by dividing each value by 255, converting them from the original range of 0–255 to a smaller range of 0–1.

This normalization helps the model train more efficiently and improves numerical stability during optimization..

B. Data Augmentation

To improve model generalization and reduce overfitting, data augmentation was applied to the training set using the following transformations:

- Rotation range: 15 degrees
- Width shift range: 0.1
- Height shift range: 0.1
- Zoom range: 0.2
- Horizontal flip: Enabled
- Fill mode: Nearest

These transformations artificially increase dataset diversity and help the model learn invariant features.

No augmentation was applied to validation or test datasets. Only rescaling was performed for those sets.

III. HYPERPARAMETERS DEFINITION

The hyperparameters were initially experimented with and systematically tuned across multiple training sessions. Different configurations were tested to improve model performance and handle class imbalance. The following values represent the final optimized hyperparameter settings used in the best-performing model.

A. Training Parameters

- Number of epochs: 20
- Batch size: 16
- Learning rate: 0.0003
- Optimizer: Adam
- Loss function: Binary Cross-Entropy
- Evaluation metric: Accuracy

These values were selected after experimentation showed improved convergence stability and validation performance.

B. Regularization Techniques

To reduce overfitting, the final model included:

- Dropout rate: 0.5 (applied to the dense layer)
- L2 regularization coefficient: 0.001

Dropout randomly disables neurons during training to prevent co-adaptation, while L2 regularization penalizes large weights to encourage simpler models.

C. Class Weighting

To address class imbalance, weighted loss was applied:

`class_weight={0:1.0, 1:1.5}`

Where:

- 0 = NORMAL
- 1 = PNEUMONIA

This configuration increases the penalty for misclassifying pneumonia cases, improving sensitivity toward the minority class.

D. Callbacks and Image Parameters

To address class imbalance, weighted loss was applied

- Early stopping patience: 5 epochs
- Reduce learning rate on plateau patience: 3 epochs
- Input image size: 224×224 pixels

These parameters were selected after experimenting with different values for batch size, learning rate, and class weighting.

IV. CNN ARCHITECTURE

A custom Convolutional Neural Network (CNN) was designed and implemented for binary classification of chest X-ray images into NORMAL and PNEUMONIA classes.

A. Detailed Architecture

1. Input LayerShape: (224, 224, 3)
2. Convolutional Block 1
 - a. Conv2D (32 filters, 3×3 , ReLU)
 - b. MaxPooling (2×2)
 - c. Batch Normalization
3. Convolutional Block 2
 - a. Conv2D (64 filters, 3×3 , ReLU)
 - b. MaxPooling (2×2)
 - c. Batch Normalization
4. Convolutional Block 3
 - a. Conv2D (128 filters, 3×3 , ReLU)
 - b. MaxPooling (2×2)
 - c. Batch Normalization
5. Flatten Layer
 - a. Converts extracted 3D feature maps into a 1D feature vector
6. Fully Connected Layer
 - a. Dense (128 neurons, ReLU activation)
 - b. L2 Regularization ($\lambda = 0.001$)
7. Dropout Layer
 - a. Dropout rate = 0.5
8. Output Layer
 - a. Dense (1 neuron)
 - b. Sigmoid activation (binary classification)

B. Design Justification

The input dimension of $224 \times 224 \times 3$ was selected to standardize image representation while preserving sufficient spatial resolution to capture clinically relevant features in chest X-ray images. Uniform input sizing ensures consistent convolutional processing across all samples.

A progressive increase in convolutional filters (32 to 64 to 128) was employed to facilitate hierarchical feature learning. The initial convolutional layers capture

low-level features such as edges and intensity gradients. Intermediate layers extract structural lung patterns, while deeper layers learn higher-level abstractions associated with pathological manifestations of pneumonia.

A 3×3 kernel size was chosen as it provides an effective balance between receptive field coverage and computational efficiency, enabling the network to capture fine-grained spatial details without excessive parameter growth.

MaxPooling (2×2) operations were incorporated to reduce spatial dimensionality, decrease computational complexity, and retain the most salient features, thereby improving translation invariance.

Batch Normalization was applied after each convolutional block to normalize intermediate activations, stabilize gradient flow, and accelerate convergence during training.

The fully connected layer comprising 128 neurons serves to integrate extracted feature representations into a compact discriminative representation for classification.

To mitigate overfitting, L2 regularization ($\lambda = 0.001$) was applied to penalize large weight magnitudes, encouraging model generalization. Additionally, Dropout (rate = 0.5) was implemented to randomly deactivate neurons during training, further enhancing robustness.

Finally, a sigmoid activation function was used in the output layer to generate probability estimates in the range [0,1], making it suitable for binary classification between NORMAL and PNEUMONIA classes.

V. EXPERIMENTAL RESULTS AND HYPERPARAMETER TUNING

To optimize model performance, multiple training experiments were conducted by systematically adjusting key hyperparameters. The objective was to improve classification accuracy, reduce overfitting, and enhance sensitivity toward pneumonia cases.

A. Training 01

TABLE I. Hyperparameters

Hyperparameters	Values
Epochs	20
Batch Size	32
Learning Rate	0.001
Dropout Rate	0.5
L2 Regularization	0.001
Early Stopping Patience	5
Reduce LR Patience	3

The evaluation results yielded an overall accuracy of 85%. The confusion matrix indicated 147 correctly classified NORMAL samples and 383 correctly classified PNEUMONIA samples. However, 87 NORMAL cases

were misclassified as PNEUMONIA, while only 7 PNEUMONIA cases were misclassified as NORMAL.

Class-wise performance analysis revealed an imbalance:

- NORMAL: Precision = 0.95, Recall = 0.63, F1-score = 0.76
- PNEUMONIA: Precision = 0.81, Recall = 0.98, F1-score = 0.89

Although pneumonia detection achieved very high recall (0.98), the recall for NORMAL cases was considerably lower (0.63). This indicates that the model exhibits a bias toward predicting the PNEUMONIA class, resulting in a high number of false positives (87 NORMAL images incorrectly classified as PNEUMONIA).

To address this issue, the batch size was reduced from 32 to 16 in the subsequent experiment. Smaller batch sizes introduce more frequent weight updates and increased gradient variability, which can improve generalization and help the model learn a more balanced decision boundary between classes.

B. Training 02 (Batch Size = 16)

In this experiment, the batch size was reduced from 32 to 16 while keeping all other hyperparameters unchanged. The objective was to evaluate the effect of smaller batch updates on model convergence and generalization performance.

The confusion matrix results indicate 215 true negatives, 19 false positives, 64 false negatives, and 326 true positives. The model achieved an overall validation accuracy of 70%.

From the classification report, the NORMAL class obtained a precision of 0.56, recall of 0.99, and F1-score of 0.71. The PNEUMONIA class achieved a precision of 0.99, recall of 0.53, and F1-score of 0.69.

Although the model maintained high precision for the pneumonia class, the recall decreased significantly, indicating that several pneumonia cases were misclassified as normal. Overall, reducing the batch size did not produce a substantial improvement in accuracy and resulted in uneven class-wise performance.

Following the observations from Training Set 02, the learning rate was adjusted to 0.0003 while maintaining the batch size of 16. The objective was to improve convergence stability and enhance generalization performance through finer gradient updates.

C. Training 03 (Learning Rate = 0.0003, Batch Size = 16)

The confusion matrix shows 203 true negatives, 31 false positives, 68 false negatives, and 322 true positives. The model achieved an overall validation accuracy of 84%, which represents a significant improvement compared to the previous training sets.

From the classification report, the NORMAL class achieved a precision of 0.75, recall of 0.87, and F1-score of 0.80. The PNEUMONIA class obtained a precision of

0.91, recall of 0.83, and F1-score of 0.87. Both classes demonstrate a more balanced precision-recall trade-off compared to earlier experiments.

The reduction in learning rate improved training stability and allowed the model to converge to a better local minimum. The decrease in false negatives and improved recall for the pneumonia class indicate enhanced sensitivity toward detecting positive cases, which is critical in medical diagnosis tasks.

D. Training 04 (Learning Rate = 0.0003, Batch Size = 16, Class Weight Adjustment)

Following the improved but still slightly imbalanced class-wise performance observed in Training Set 03, class weights were introduced as:

`class_weight = {0: 1.0, 1: 1.2}`

where class 0 corresponds to NORMAL and class 1 corresponds to PNEUMONIA.

The purpose of increasing the weight of the pneumonia class was to penalize misclassification of pneumonia samples more heavily during training. By assigning a higher weight to class 1, the loss function emphasizes reducing false negatives, thereby improving recall for pneumonia cases. This adjustment aims to further enhance model sensitivity toward the minority or clinically critical class without significantly compromising overall accuracy.

The confusion matrix shows 219 true negatives, 15 false positives, 79 false negatives, and 311 true positives. The overall validation accuracy increased to 85%.

From the classification report, the NORMAL class achieved a precision of 0.73, recall of 0.94, and F1-score of 0.82. The PNEUMONIA class obtained a precision of 0.95, recall of 0.80, and F1-score of 0.87. The macro-average F1-score improved to 0.85, and the ROC-AUC score reached 0.9525, indicating strong separability between the two classes.

Compared to Training 03, the introduction of class weighting improved overall accuracy and reduced false positives significantly (from 31 to 15). However, the number of false negatives slightly increased (from 68 to 79), indicating that while precision for pneumonia improved, recall did not improve proportionally.

Although the model achieved better overall discrimination performance, further improvement in pneumonia recall remained desirable due to its clinical importance. Therefore, the class weight was further adjusted.

E. Training 05 (Learning Rate = 0.0003, Batch Size = 16, Class Weight Adjustment)

The class weight was further adjusted to:

`class_weight = {0: 1.0, 1: 1.4}`

to place a stronger penalty on misclassified pneumonia cases and further enhance model sensitivity toward the minority class.

The confusion matrix shows 215 true negatives, 19 false positives, 64 false negatives, and 326 true positives. The overall validation accuracy improved to 87%, representing the highest performance achieved among all previous training configurations.

From the classification report, the NORMAL class achieved a precision of 0.77, recall of 0.92, and F1-score of 0.84. The PNEUMONIA class obtained a precision of 0.94, recall of 0.84, and F1-score of 0.89. The macro-average F1-score increased to 0.86, while the weighted-average F1-score reached 0.87. The ROC-AUC score further improved to approximately 0.954, indicating strong discriminatory capability.

Compared to Training 04, increasing the pneumonia class weight reduced false negatives (from 79 to 64) and improved recall for pneumonia (from 0.80 to 0.84), while maintaining high precision. This demonstrates that stronger class penalization successfully enhanced model sensitivity without significantly degrading performance on the NORMAL class.

Given the continued improvement in pneumonia detection performance, the class weight for class 1 was further increased to 1.5.

F. Training 06 (Learning Rate = 0.0003, Batch Size = 16, Class Weight Adjustment)

The class weight was further adjusted to:

`class_weight = {0: 1.0, 1: 1.5}`

The resulting confusion matrix indicates 177 true negatives, 57 false positives, 21 false negatives, and 369 true positives. The overall test accuracy improved to 88%, representing the highest accuracy achieved among all conducted experiments.

From the classification report, the NORMAL class achieved a precision of 0.89, recall of 0.76, and F1-score of 0.82. The PNEUMONIA class obtained a precision of 0.87, recall of 0.95, and F1-score of 0.90. The macro-average F1-score reached 0.86, while the weighted-average F1-score was 0.87. Furthermore, the ROC-AUC score was 0.9500, indicating strong discriminative capability of the proposed CNN model.

G. Training 07 (Learning Rate = 0.0003, Batch Size = 16, Class Weight Adjustment, Dropout Rate = 0.4)

The final model configuration demonstrated strong and balanced performance on the test dataset. Based on the confusion matrix, the model correctly classified 200 normal cases and 352 pneumonia cases, while misclassifying 34 normal images as pneumonia (false positives) and 38 pneumonia images as normal (false negatives). Out of a total of 624 test samples, 552 were correctly predicted, resulting in an overall classification accuracy of 88%. These results indicate that the proposed CNN architecture generalizes well to unseen data.

For the NORMAL class, the model achieved a precision of 0.84, recall of 0.85, and an F1-score of 0.85, showing balanced performance in identifying healthy cases. For the PNEUMONIA class, the precision reached 0.91, recall 0.90, and F1-score 0.91, demonstrating slightly stronger performance in detecting infected cases.

The high recall for pneumonia (90%) is particularly significant in a medical diagnosis context, as it indicates that the majority of positive cases were correctly identified. The macro-average and weighted-average F1-scores were both 0.88, suggesting consistent performance across classes without severe bias toward the majority class.

Additionally, the model achieved an AUC-ROC score of 0.9498, indicating excellent discriminative ability between normal and pneumonia cases across varying decision thresholds. Although a small number of false positives and false negatives remain, the error distribution is relatively balanced, reflecting stable learning behavior and effective generalization. Overall, the final model achieves a strong trade-off between sensitivity and specificity, making it suitable for automated pneumonia detection from chest X-ray images.

VI. COMPARISON WITH PRE-TRAINED MODELS

A comparative analysis was conducted with two widely used deep learning architectures: ResNet50 and VGG16. All models were trained and evaluated on the same dataset.

A. ResNet50

Confusion Matrix

TABLE II. ResNet50 Confusion Matrix

	Predicted Normal	Predicted Pneumonia
Actual Normal	227	7
Actual Pneumonia	146	244

Classification Metrics

TABLE III. ResNet50 Classification Metrics

Class	Precision	Recall	F1-Score
NORMAL	0.61	0.97	0.75
PNEUMONIA	0.97	0.63	0.76

Explanation

When compared with ResNet50, the Custom CNN demonstrated substantially superior performance. ResNet50 achieved an overall accuracy of approximately 75%, with a strong recall for the NORMAL class (0.97) but a relatively low recall for the PNEUMONIA class (0.63). This indicates that while ResNet50 was highly effective at identifying normal cases, it failed to detect a significant number of pneumonia cases, resulting in a large number of false negatives.

In medical diagnosis, such false negatives are critical because missed pneumonia cases may delay treatment and increase health risks. Therefore, despite its strong performance in identifying normal cases, the imbalance in class-wise recall limits the practical reliability of ResNet50 for clinical screening tasks.

In contrast, the Custom CNN achieved an accuracy of 88% and significantly improved pneumonia recall to 0.90. The number of false negatives was drastically reduced, and the performance across both classes became more balanced. The F1-scores for NORMAL (0.85) and PNEUMONIA (0.91) further demonstrate that the Custom CNN learned a more stable and clinically reliable decision boundary.

These results indicate that the tuned Custom CNN is more suitable for this specific medical imaging task than ResNet50, despite ResNet50's deeper and more complex architecture.

B. VGG16

Confusion Matrix

TABLE IV. VGG16 Confusion Matrix

	Predicted Normal	Predicted Pneumonia
Actual Normal	217	17
Actual Pneumonia	41	349

Classification Metrics

TABLE V. VGG16 Classification Metrics

Class	Precision	Recall	F1-Score
NORMAL	0.84	0.93	0.88
PNEUMONIA	0.95	0.89	0.92

Explanation

When comparing the Custom CNN with VGG16, the difference in performance was small. VGG16 achieved an overall accuracy of about 91%, which is slightly higher than the Custom CNN's 88%. It also showed strong and balanced results for both classes, with F1-scores of 0.88 for NORMAL and 0.92 for PNEUMONIA.

The recall for pneumonia in VGG16 was 0.89, which is very close to the Custom CNN's recall of 0.90. This means both models were almost equally good at detecting pneumonia cases. Since identifying positive cases is very important in medical diagnosis, this shows that the Custom CNN performs nearly as well as VGG16 in this area.

However, VGG16 is a deeper and more complex model, which requires more computational power and training time. The Custom CNN, on the other hand, was specially designed and tuned for this dataset using hyperparameter tuning, class weighting, and regularization. Even though it is simpler, it still achieved competitive results, including a high AUC-ROC score of 0.9498. This shows that the Custom CNN is efficient and performs very well without needing a very large or complex architecture.

VII. SAMPLE DATASET IMAGES

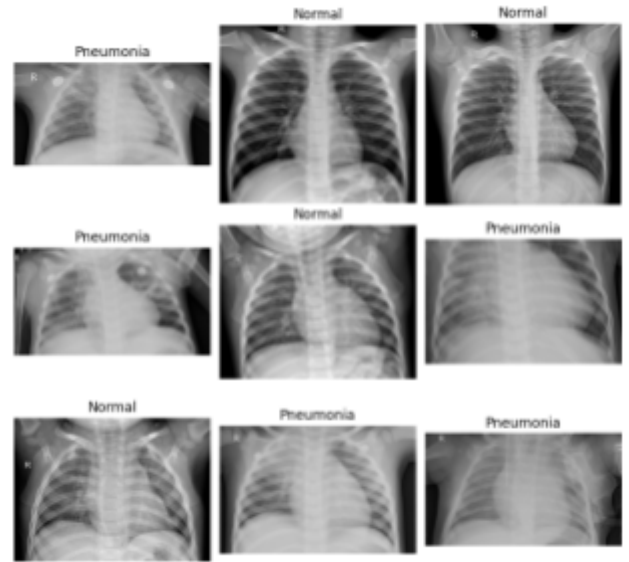


Fig.1. Sample Dataset Images

VIII. PREDICTION RESULTS

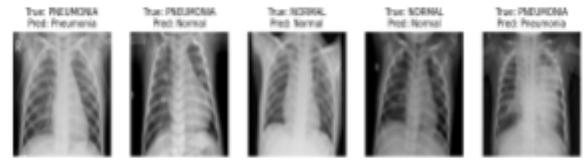


Fig.2. Prediction Results

IX. TRAINING VS VALIDATION LOSS AND ACCURACY CURVES

A. Training vs. Validation Loss

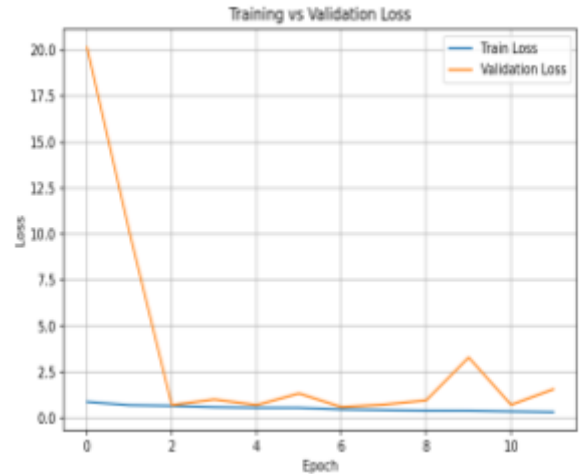


Fig.3. Training Vs Validation Loss

Training accuracy increases steadily to around 95%, showing the model learns training patterns consistently.

Validation accuracy is highly unstable (about 50% to 94%), indicating poor generalization and likely overfitting/high variance.

B. Training vs. Validation Accuracy



Fig.4. Training Vs Accuracy Curve

Training loss gradually decreases, which suggests smooth optimization on the training set.

Validation loss is erratic with a very large initial spike and later fluctuations, confirming unstable validation behavior and unreliable convergence.